Practical Problem 3: Decision Tree using ID3 Algorithm

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
# Load dataset
df = pd.read_excel('/content/play_tennis.csv.xlsx')
print(df)
# Convert text values into numbers
encoders = {}
for col in df.columns:
    encoders[col] = LabelEncoder()
    df[col] = encoders[col].fit_transform(df[col])
# Separate input and target
X = df.drop('Target', axis=1)
y = df['Target']

# Build ID3 Decision Tree
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)

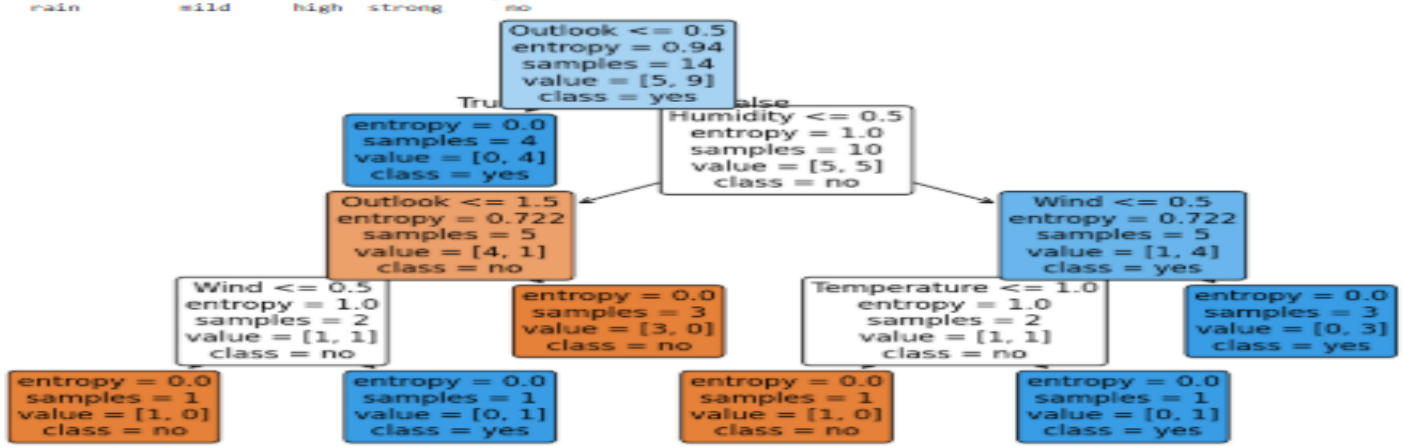
# Display decision tree
plt.figure(figsize=(12, 7))
plot_tree(model, feature_names=X.columns,
          class_names=encoders['Target'].classes_,
          filled=True, rounded=True)
plt.show()

# Classify a new sample
sample = pd.DataFrame([{
    'Outlook': 'sunny',
    'Temperature': 'cool',
    'Humidity': 'normal',
    'Wind': 'strong'
}])

# Encode sample using the existing encoders
for col in sample.columns:
    sample[col] = encoders[col].transform(sample[col])

prediction = model.predict(sample)[0]
print("New Sample: sunny, cool, normal, strong")
print("Predicted Target:", encoders['Target'].inverse_transform([prediction])[0])
```

```
      Outlook Temperature Humidity    Wind Target
0       sunny         hot     high    weak     no
1       sunny         hot     high  strong     no
2    overcast         hot     high    weak    yes
3        rain        mild     high    weak    yes
4        rain        cool   normal    weak    yes
5        rain        cool   normal  strong     no
6    overcast        cool   normal  strong    yes
7       sunny        mild     high    weak     no
8       sunny        cool   normal    weak    yes
9        rain        mild   normal    weak    yes
10      sunny        mild   normal  strong    yes
11   overcast        mild     high  strong    yes
12   overcast         hot   normal    weak    yes
13       rain        mild     high  strong     no
```




```
New Sample: sunny, cool, normal, strong
Predicted Target: no
```